

WELKOM, South Africa

WMO: 683450

Lat: 27.995S

Lon: 26.666E

Elev: 1344

StdP: 86.19

Time Zone: 2.00 (E02)

Period: 98-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	0.2	1.7	-12.8	1.5	16.1	-10.9	1.7	16.0	10.1	16.5	9.2	15.4	2.6	120	0.446

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	12.3	34.4	16.1	33.1	16.1	31.9	16.0	20.6	27.3	20.0	26.6	19.5	26.0	4.9	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.9	16.1	22.5	18.2	15.4	22.1	17.6	14.8	21.9	66.2	27.3	63.8	26.5	61.8	26.0	27.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.0	8.9	7.9	DB	-2.2	36.9	1.4	1.8	-3.2	38.2	-4.0	39.3	-4.8	40.3	-5.8	41.6	(4)
(5)			WB	-4.1	21.6	1.3	2.3	-5.0	23.3	-5.7	24.6	-6.4	25.9	-7.3	27.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	18.4	23.8	23.4	21.6	18.1	14.5	11.4	11.2	14.0	18.1	20.4	22.0	23.1	(6)	
	DBStd	5.31	2.34	2.14	2.19	2.67	2.91	2.65	2.71	3.81	3.58	3.23	3.24	2.49	(7)	
	HDD10.0	51	0	0	0	0	4	16	19	11	1	0	0	0	(8)	
	HDD18.3	804	1	1	2	34	120	207	222	142	48	17	9	2	(9)	
	CDD10.0	3131	427	376	358	244	145	59	56	134	243	322	358	408	(10)	
	CDD18.3	842	170	143	102	29	2	0	0	6	39	81	118	151	(11)	
	CDH23.3	8388	1517	1221	851	291	71	4	2	122	581	967	1310	1453	(12)	
(13)	CDH26.7	3213	645	482	266	41	6	0	0	18	184	375	573	623	(13)	
(14)	Wind	WSAvg	3.7	4.0	3.7	3.5	3.1	2.8	3.0	3.1	3.9	4.2	4.4	4.6	4.5	(14)
(15) Precipitation	PrecAvg	501	75	57	78	38	23	7	6	11	12	47	68	83	(15)	
	PrecMax	776	170	162	139	136	155	30	44	56	59	137	178	156	(16)	
	PrecMin	284	24	13	13	1	0	1	0	0	1	7	3	21	(17)	
	PrecStd	150	39	33	43	35	35	8	11	17	13	36	47	47	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	36.3	34.6	33.9	29.8	27.4	23.6	23.4	28.7	32.2	34.1	35.3	36.1	(19)	
		MCWB	16.8	16.4	17.4	15.2	12.9	10.4	9.9	12.7	13.3	14.1	16.0	15.8	(20)	
	2%	DB	33.8	32.9	31.2	27.8	25.2	21.7	21.8	26.4	30.5	32.4	33.5	33.7	(21)	
		MCWB	17.4	17.2	16.6	15.5	12.1	9.7	9.6	11.5	13.0	14.3	15.3	16.2	(22)	
	5%	DB	32.0	31.3	29.5	26.5	23.7	20.2	20.2	24.5	28.9	30.8	31.9	32.0	(23)	
		MCWB	17.2	17.9	16.8	15.2	11.6	9.4	9.2	10.9	12.3	13.9	15.3	16.4	(24)	
	10%	DB	30.3	29.8	28.0	25.0	22.1	18.9	18.7	22.5	27.0	28.9	30.3	30.2	(25)	
		MCWB	17.6	18.3	16.9	14.5	11.4	8.9	8.6	10.1	11.9	13.8	15.2	16.5	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	21.2	21.4	21.1	18.5	15.1	12.7	12.1	14.0	15.6	17.8	19.4	20.6	(27)	
		MCDB	27.9	28.2	26.6	24.7	22.4	19.3	19.1	24.7	24.9	25.9	27.7	27.6	(28)	
	2%	WB	20.3	20.7	19.9	17.5	13.8	11.5	10.7	12.7	14.5	17.0	18.5	19.7	(29)	
		MCDB	27.0	27.5	25.9	23.8	21.4	18.7	19.2	24.0	25.1	25.5	26.8	26.7	(30)	
	5%	WB	19.7	20.1	19.1	16.7	13.0	10.5	9.9	11.8	13.6	16.2	17.7	19.1	(31)	
		MCDB	26.3	26.8	25.3	22.9	20.6	17.8	18.8	22.2	25.0	24.6	26.1	26.0	(32)	
	10%	WB	19.2	19.5	18.2	16.0	12.2	9.6	8.9	10.8	12.9	15.5	17.1	18.4	(33)	
		MCDB	25.6	25.9	24.6	22.6	20.0	16.9	17.1	20.3	24.2	23.7	25.5	25.5	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	12.3	11.8	12.4	12.8	15.0	15.0	14.7	15.5	16.0	14.8	14.6	13.3	(35)	
		MCDBR	15.1	14.6	14.8	14.6	16.7	16.4	16.3	17.0	17.8	17.2	16.6	16.1	(36)	
	5% WB	MCWBR	4.1	4.3	4.8	5.3	6.7	7.5	7.5	7.3	7.0	6.0	5.0	4.7	(37)	
		MCWBR	11.5	11.2	11.9	11.8	13.8	14.0	14.4	15.3	15.5	13.4	13.3	12.0	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.361	0.364	0.351	0.313	0.284	0.269	0.268	0.321	0.360	0.384	0.339	0.346	(40)	
		taud	2.375	2.371	2.381	2.467	2.523	2.532	2.514	2.350	2.237	2.212	2.385	2.404	(41)	
	Ebn at Noon		981	960	941	932	919	913	928	908	916	929	996	998	(42)	
		Edh at Noon	130	127	119	99	84	79	83	109	135	148	128	127	(43)	
(44) All-Sky Solar Radiation	RadAvg		7.29	6.63	5.84	4.84	4.17	3.75	4.10	4.93	6.03	6.73	7.40	7.55	(44)	
	RadStd		0.65	0.63	0.38	0.33	0.20	0.17	0.12	0.25	0.27	0.39	0.45	0.40	(45)	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (5 neighbors)		+0.65	+0.76	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page